

Econ H191 Final Paper – Detailed Outline and Transition Plan

Part I – Detailed Outline for the Econ H191 Final Paper

This outline is meant to be **Overleaf-ready**: each numbered section corresponds to a major section in the final paper, with suggested subsections, core content, and notes on figures/tables.

Front Matter (unnumbered)

Title Page - Title (working): *Capital Deepening and Labor Productivity Under Port Reform: Evidence from Israel's 2021–2023 Port Reforms*. - Your name, department(s), advisor, course (Econ H191 – Honors Thesis), date.

Abstract ($\approx$ 200–250 words) - 1–2 sentences: context and problem (Israel's productivity gap, missing capital, port inefficiency). - 1–2 sentences: data (terminal $\times$ month panel, new capital series, labor data) and methods (event-study, NYT design, elasticity model). - 2–3 sentences: key results for LP and K/L, including approximate magnitudes and timing. - 1–2 sentences: mediation-style decomposition (share of LP change consistent with K/L) + main takeaway.

Acknowledgments (short, optional) - Thank advisor(s), faculty you consulted, data providers (Ports Company/Ministry), and anyone who helped.

Table of Contents; List of Figures; List of Tables - Let Overleaf generate these, but think about a clean label system for figures/tables.

1 Introduction ($\approx$ 4–6 pages)

1.1 Motivation and macro context - Israel's persistent output-per-worker gap relative to similar rich countries. - Emphasize the macro finding that low *physical capital per worker* (especially structures) is a key driver of this gap. - Explain why infrastructure-heavy, non-tradable sectors (like ports) are central to the "missing capital" story. - One paragraph on why ports matter specifically for trade costs, competition, and the cost-of-living debates.

1.2 Israeli ports as a case study - Brief description of Haifa and Ashdod as the two main Mediterranean container gateways. - Historical issues: state-owned duopoly, powerful unions, long ship waits, underinvestment, and bottlenecks. - Summarize in 1 paragraph how these ports became a symbol of structural inefficiency and high import costs.

1.3 Research questions - Q1: How did competition from new private terminals and the privatization of the Haifa legacy operator affect *terminal-level labor productivity* (LP)? - Q2: How did these reforms affect *capital*

deepening (K/L) at the same terminals? - Q3: To what extent are observed LP gains consistent with changes in K/L, versus other channels (e.g., work practices, utilization, management)?

1.4 Approach and main results (preview) - One paragraph: data and measurement - New monthly and quarterly series for output, labor hours, and capital at port/terminal level. - Construction of capital from PPE flows + project-level data (Path B) and labor productivity from throughput and hours. - One paragraph: empirical strategy - NYT event-study models (Models 1A and 1B) for reforms → LP and reforms → K/L. - Elasticity model (Model 2) relating LP to K/L. - Mediation-style decomposition that combines these pieces. - One paragraph: headline results (filled once estimates exist) - Direction and rough magnitude of reform effects on LP and K/L. - Approximate share of LP change “accounted for” by K/L.

1.5 Contributions - Contribution 1: First high-frequency causal estimates of Israel's 2021–2023 port reforms on terminal-level LP. - Contribution 2: A method for constructing monthly capital stocks and K/L in a port-sector setting using firm-level PPE and big-project data. - Contribution 3: A mechanism accounting exercise that quantifies how much of reform-induced LP changes are consistent with capital deepening in a setting with overlapping reforms and strong unions.

1.6 Roadmap - A short paragraph explaining the structure: - Section 2: Institutional background. - Section 3: Related literature and conceptual framework. - Section 4: Data and measurement. - Section 5: Econometric strategy. - Section 6: Results. - Section 7: Discussion. - Section 8: Conclusion.

2 Institutional Background: Israel's Ports and the Competition Reform (≈ 5–7 pages)

2.1 The Israeli port system pre-reform - Governance before 2003: Israel Ports Authority as an integrated monopoly. - Basic port functions (marine vs landside), who owned what (infrastructure vs operations). - Highlight: long waits, low comparative productivity, strong unions, and lack of competition.

2.2 Corporatization and the creation of IPC (2003–2005) - Split into Haifa Port Company (HPC), Ashdod Port Company (APC), and Eilat; creation of Israel Ports Company (IPC) as landlord/regulator. - How ownership, investment responsibilities, and decision-making changed. - Limited gains: some service improvements but persistent structural issues and concentrated market power.

2.3 The 2011 cost-of-living crisis and policy pivot - Summarize the social protests and the Trajtenberg Committee's diagnosis. - Explicit link between port inefficiency, lack of intra-port competition, and high consumer prices. - Recommendation to build new deep-water terminals in Haifa and Ashdod to introduce competition.

2.4 Deep-water competition projects: Bayport and HCT - Basic technical description: bay location, depth, capacity, operator (SIPG, TIL). - Chronology: tenders, building phase, commissioning, ramp-up. - Describe which legacy terminals they compete with and how volumes can shift between terminals.

2.5 Privatization of Haifa legacy - Process and timeline of selling Haifa Port Company's operations via concession. - Changes in ownership and possible implications for investment incentives and labor relations.

2.6 How reforms may affect LP and K/L - Intuitive channels: - Competition → pressure to invest in cranes/berths → higher K/L. - Competition → operational improvements, better shift management, less downtime. - Privatization → governance, incentives, possible renegotiation of union power. - Explain that the empirical design will separately track impacts on LP and K/L, and interpret “what is left over” after accounting for K/L as other mechanisms.

3 Related Literature and Conceptual Framework (≈ 5–7 pages)

3.1 Macro evidence on Israel's productivity and capital deepening - Summarize the macro literature diagnosing Israel's productivity gap. - Emphasize findings on low K/L, especially in structures and infrastructure-intensive sectors. - Link explicitly to the port sector as a natural micro setting to observe capital deepening.

3.2 Port reforms, competition, and productivity worldwide - Short review of empirical studies on port reforms in other countries. - Highlight main patterns: - Intra-port competition is more strongly associated with productivity gains than privatization alone. - Capital deepening and technology upgrades are often mentioned as mechanisms but rarely measured explicitly.

3.3 Methodological background: DiD, event-studies, and mechanisms - Briefly describe modern staggered-adoption DiD issues and NYT/event-study solutions. - Distinguish between: - Event-study estimates of causal reform effects on outcomes. - Mediation / mechanism analysis which uses other regressions or structural relationships.

3.4 Conceptual production framework - Introduce a simple production function at the terminal level: $Y = A K^\alpha L^{1-\alpha}$, define $LP = Y/L$ and K/L . - Show how $\log(LP)$ decomposes into $\log(A) + \alpha \log(K/L)$. - Discuss: - How reforms can affect A (efficiency, utilization, management) and K/L (investment, labor changes). - Why we treat K/L as a *measured mediator/outcome*, not an instrument. - Explain that Model 2's elasticity is the empirical counterpart of α , but will be interpreted cautiously.

4 Data and Measurement (≈ 7–10 pages)

4.1 Overview of data sources and sample construction - List main sources: - Port throughput series (monthly, quarterly). - KPI series (TEU/hour, waiting/berth/stay times). - Labor-hours datasets from ports and/or ministries. - Financial statements and Note-8 PPE data. - Project-level data on major capital works. - Define units of analysis: - Pre-competition: port×month. - Post-competition: terminal×month or terminal×quarter. - Define sample period, coverage, and any missing segments.

4.2 Output measurement - Describe how TEU and tons series are collected and cleaned. - Explain how you aggregate to port/terminal, align time frequencies, and treat special cargo types. - Describe any mix-adjustment (tons/TEU) used to make LP comparable.

4.3 Labor measurement and LP construction - Final labor measure: - Describe how you obtain monthly labor hours per port/terminal. - Data cleaning: merging across sources, removing outliers, handling missing months. - Construct LP: - $LP = \text{output} / \text{labor hours}$, at port×month or terminal×month. - Define $\log(LP)$ as

primary outcome variable. - Proxies and robustness: - Describe the earlier KPI-based LP proxy and how it will be used as a robustness or backfill for early periods. - Discuss potential biases from proxies vs true hours.

4.4 Capital measurement: Path B (PPE + big projects)

4.4.1 Annual PPE backbone - Describe the financial statements used (Haifa Port Company, possibly others). - Key variables: - Net PPE at year-end, additions, disposals, depreciation. - Explain the capital deflator and how you convert nominal PPE and flows to real values.

4.4.2 Perpetual Inventory Method (annual K) - Explain the PIM logic: $K_t = (1-\delta)K_{t-1} + I_t$ in real terms. - Specify chosen depreciation structure (e.g., geometric $\delta \approx 6\%$ per year; note planned sensitivity checks). - Show how you back-calculate an initial K_0 consistent with observed net PPE.

4.4.3 Big-project overlay - Describe the project list: STS cranes, yard cranes, berth deepening, Kishon projects, seafront relocation, Sea Department sale, etc. - Explain how each project is: - Assigned to a commissioning year/month. - Given a real cost and asset life. - Categorized by asset class (crane, berth, yard, etc.). - Explain how you ensure consistency: big-project costs never exceed annual additions; disposal projects (like the Sea Department sale) are treated with negative signs.

4.4.4 Monthly capital series - Describe the interpolation from annual K to monthly K: - Basic method (e.g., linear or Denton-like, respecting annual totals). - Superimposing step changes from project commissioning. - Mention how you map aggregate K to specific terminals (legacy vs new) based on berth assignment and project location.

4.4.5 From K to K/L - Define K/L at the terminal-month level and its log. - Discuss how you handle months with zero or missing labor (if any) and how you treat pre-competition periods.

4.5 Reform timing, event-time clocks, and shocks - Precisely define the three reforms: - Haifa competition entry (Bayport opening date). - Ashdod competition entry (HCT opening date). - Haifa privatization (closing date of concession sale). - Define event-time variables for each reform and the binning scheme (e.g., individual months vs combined windows). - List exogenous shocks to control for (e.g., COVID, war periods) and how they enter regressions.

4.6 Descriptive statistics and graphs - Plan a set of descriptive tables: - Summary stats for LP and K/L by terminal and period (pre/post). - Summary of capital stock levels and growth. - Plan a small set of figures: - Time series of LP and K/L by terminal, with reform dates marked. - Capital stock evolution for Haifa legacy, showing project jumps.

5 Econometric Strategy ($\approx$ 6–8 pages)

5.1 Identification: what we want and where identification comes from - Clarify the parameters of interest: - Dynamic effects of reforms on LP and on K/L (Models 1A and 1B). - Elasticity of LP with respect to K/L (Model 2). - Mediation-style decomposition: how much of the LP effect is consistent with K/L. - Explain why simple two-way FE DiD is problematic in staggered settings and why NYT/event-study is preferred.

5.2 Model 1A – NYT event-study: Reform → $\ln(LP)$ - Write the regression equation clearly with notation. - Define treatment, event-time bins, and the NYT control set for each reform. - Explain how you will: - Implement FE (terminal and month), clustering, and potential port-specific trends. - Summarize dynamic effects via window averages (e.g., 1–12, 13–24, full post, pre-trend window). - State the identification assumption: parallel trends in $\ln(LP)$ between treated and NYT comparison group in absence of reform.

5.3 Model 1B – NYT event-study: Reform → $\ln(K/L)$ - Clone the specification with $\ln(K/L)$ as the outcome. - Emphasize interpretive focus: - This traces capital deepening response to reforms with the same design and comparison group structure. - It is not an IV first stage, but a separate outcome.

5.4 Model 2 – LP elasticity with respect to K/L - Write the baseline regression with $\ln(LP)$ on $\ln(K/L)$, including FE. - Discuss: - Contemporaneous vs lagged K/L. - Why this is an elasticity estimate, not full causal effect from an exogenous capital shock. - Mention planned robustness: - Alternative specifications (first differences, alternative controls). - Cross-check against implied capital shares from accounts.

5.5 Mediation-style decomposition - Explain in words how Models 1A, 1B, and 2 are combined: - Use event-study window averages for LP and K/L. - Use the elasticity estimate to translate $\Delta \ln(K/L)$ into an implied $\Delta \ln(LP)$. - Compare implied $\Delta \ln(LP)$ to the total event-study $\Delta \ln(LP)$. - Make clear that this is *mechanism accounting*, not Dippel–Ferrara IV mediation.

5.6 Robustness and sensitivity strategy - Pre-commit to robustness checks: - Alternative binning and event windows. - Stacked event-studies when pooling multiple reforms. - Alternative labor series (different definitions and allocations). - Alternative capital variants (different depreciation rates, treatment of revaluations, project subsets). - Excluding periods with extreme shocks (e.g., war months) as a check.

6 Results ($\approx$ 8–12 pages once filled)

6.1 Descriptive evolution of LP, K/L, and throughput - Present and discuss key descriptive figures and tables. - Highlight patterns around reform dates qualitatively (before invoking causal estimates).

6.2 Event-study results for LP (Model 1A) - Show main ES plots by reform and terminal type (entrant/legacy). - Provide window-average estimates in tables (short-run, medium-run, full post). - Comment on pre-trends, timing, and magnitude.

6.3 Event-study results for K/L (Model 1B) - Show parallel ES plots for K/L. - Compare timing of K/L changes to LP changes: - Does K/L move first, together, or later? - Are capital responses large/small relative to LP?

6.4 Elasticity estimates (Model 2) - Present main elasticity estimate and key variants in a compact table. - Discuss its plausibility vs capital-share benchmarks and macro literature.

6.5 Mediation-style decomposition - Present a summary table: - For each reform and window: total LP effect, K/L effect, implied LP-from-K/L, residual component, and share accounted-for. - Interpret patterns: - Where does capital deepening explain a lot? Where does it explain little?

6.6 Robustness checks - Summarize robustness results (full tables and plots can go in appendices). - Emphasize which findings are stable and where results are sensitive to assumptions about K or L.

7 Discussion ($\approx$ 4–6 pages)

7.1 Interpreting magnitudes and channels - Place the estimated LP changes in context (e.g., relative to baseline productivity, relative to macro gaps). - Consider the scale of capital changes: are reforms driving big or modest increases in K/L?

7.2 Capital deepening vs other mechanisms - Interpret residual LP changes not explained by K/L. - Discuss potential channels: - Work rules and union agreements. - Management and process changes. - Utilization and scheduling improvements. - Technology and IT systems beyond what's captured in K.

7.3 Connection to broader literatures - Connect your findings to: - Macro work on Israel's missing capital. - International evidence on port reforms and privatization. - Broader debates on competition, investment, and productivity in non-tradables.

7.4 Policy implications and external validity - What do your results suggest about: - The effectiveness of introducing intra-port competition? - The role of privatization vs competition in driving investment and productivity? - Discuss whether your findings likely generalize to other infrastructure sectors and countries.

8 Conclusion ($\approx$ 2–3 pages)

8.1 Summary of findings - Concise recap of: - Reform effects on LP. - Reform effects on K/L. - Contribution of K/L to LP changes.

8.2 Limitations - Data limitations (missing months, level of detail on capital and labor, small N of treated units). - Potential biases (endogeneity, measurement error).

8.3 Future research - Extending time horizon as more post-reform data accumulate. - Incorporating more detailed micro data (ship-level waiting times, price/fee data, customer-level outcomes). - Cross-country comparisons of similar reforms.

9 References

- Standard references section; follow a consistent citation style (e.g., Chicago author-year).
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10 Appendices

Appendix A – Data Appendix & Variable Definitions - Detailed tables of variables, units, sources, and construction steps. - Summary of data cleaning procedures.

Appendix B – Institutional Details and Chronology - Long-form narrative of port governance and reforms. - Maps, diagrams, and timelines.

Appendix C – Additional Figures and Tables - Full event-time coefficient tables. - Additional descriptive plots.

Appendix D – Robustness and Placebo Checks - Stacked ES results, alternative K/L definitions, alternative labor measures, shock-exclusion checks.

Part II – Strategy for Transitioning from Proposal to Final Paper

This part lays out a concrete plan to move from the existing proposal and redesign memo to the new final-paper structure.

A. Map Existing Material to the New Outline

1. Introduction & Motivation

2. Reuse: The first section of the proposal already has strong motivation, context, and literature references.
3. Task: Update language to reflect the new role of K/L (measured outcome, not IV mediator) and new econometric design.

4. Institutional Background

5. Reuse: The Research Dossier and parts of the proposal's background can be condensed and reorganized into Section 2.
6. Task: Distill into a clear chronological narrative, avoid duplications, and keep empirical design references out of this section.

7. Literature Review

8. Reuse: The proposal's literature review is well-developed and can be adapted into Section 3.
9. Task: Reorganize into subsections (Israel macro, port reforms, methodology), and add a short conceptual production-function discussion at the end.

10. Data & Measurement

11. Reuse: The proposal's tables and descriptions of throughput series, KPI-based LP, and labor proxies.
12. Task: Integrate with updated capital and labor pipelines from the data/code work, and reframe KPIs as proxies/robustness rather than core definitions.

13. Econometric Design

14. Reuse: The existing NYT event-study description for LP (Model 1A) is still valid.

15. Replace: The entire IV-mediated Model 2 section from the proposal must be replaced with the updated three-model structure (Models 1A, 1B, 2 + mediation accounting).

16. Initial Results

17. Reuse: The event-study tables and figures templates in the proposal's "overview & initial results" section.

18. Task: Update to reflect final data and new K/L outcomes; integrate with final Results section structure.

B. Rewriting Plan by Section

Step 1 – Lock the Introduction (Section 1) - Start from the existing abstract and intro paragraphs in the proposal. - Rewrite to: - Emphasize measured K/L and LP as twin outcomes. - Clearly state the new econometric design (no IV mediation; instead, event-studies + elasticity + mediation accounting). - Align contributions with capital measurement and mechanism accounting. - Deliverable: A polished 4–6 page introduction ready for Overleaf.

Step 2 – Build Section 2 (Background) from the Dossier - Use the Research Dossier as the main source. - Steps: - Extract and condense key background paragraphs (pre-reform structure, corporatization, Trajtenberg reforms, Bayport/HCT projects, privatization). - Ensure the narrative is chronological and clean. - Avoid including data detail or equations; those belong in later sections. - Deliverable: A coherent background section with subsections 2.1–2.6.

Step 3 – Reorganize the Literature Review (Section 3) - Take the proposal's literature section and split it into clear subsections: macro Israel, port reforms, methodology. - Add a short conceptual production-function subsection at the end. - Make sure every cited paper has a clear role (motivation vs comparison vs method).

Step 4 – Upgrade Data & Measurement (Section 4) - Combine: - Proposal's "Observed Data and building proxies" section. - Redesign memo's descriptions of Path B capital and the new LP/L construction. - Steps: - Elevate capital measurement (subsection 4.4) to a centerpiece, using the Path B narrative. - Clarify final labor data versus proxy solutions; be honest about limitations and the transition. - Define all variables formally (LP, K, L, K/L, event-time variables) in this section. - Deliverable: A detailed data section that sets up all subsequent econometric work.

Step 5 – Rewrite Econometric Strategy (Section 5) - Use the redesign memo as the main template for this section. - Steps: - Replace all IV mediation language with the updated three-model structure. - Write each model's equation clearly and consistently with the notation used in Section 4. - Add a mediation-style decomposition subsection that is clear and notationally clean. - Pre-commit robustness checks and explain why they matter.

Step 6 – Results and Robustness (Sections 6 & Appendices) - Start with the event-study plots already sketched in the proposal. - Steps: - Once final data are in place, produce the key ES figures and window-average tables for LP and K/L. - Estimate the elasticity model and construct the mediation table. - Organize robustness checks and place detailed tables in the appendices. - Deliverable: A results section that flows from descriptive → event-study → elasticity → mediation → robustness.

Step 7 – Discussion and Conclusion (Sections 7–8) - Once results are known, draft a discussion that: - Interprets magnitudes relative to macro constraints and the literature. - Distinguishes capital deepening from other mechanisms. - Extracts policy implications. - Then write the conclusion, summarizing contributions and pointing to future work.

C. Workflow and Version Control Strategy

1. Create a clean Overleaf project

2. Name: `Econ_H191_Final_Paper`.

3. Start from a standard econ article or thesis template with sections matching the outline above.

4. Modular rewriting

5. Work section-by-section, pasting in relevant proposal text and then editing heavily.

6. Maintain a clear comment mark (`% OLD PROPOSAL TEXT BELOW`) where you temporarily keep legacy text for reference.

7. Remove or archive legacy blocks once they are fully integrated.

8. File structure in Overleaf

9. `main.tex` with section inputs:

◦ `intro.tex`, `background.tex`, `literature.tex`, `data.tex`, `design.tex`,
`results.tex`, `discussion.tex`, `conclusion.tex`, `appendix.tex`.

10. Keep tables/figures in dedicated `.tex` or `.csv` / `.tsv` files if you're using external inputs.

11. Synchronize with your local repo

12. Keep your data and code in the local `THESIS/` repo.

13. Periodically export final tables and figures from Python/Stata into `Output/` and then into Overleaf.

14. Iterative refinement

15. After a first full draft exists, do at least two passes:

- Structural pass: check that each section logically flows to the next and reflects the updated design.

- Style pass: tighten language, unify notation, and ensure consistent terminology ("LP", "K/L", "legacy", "entrant", etc.).

16. Advisor-facing milestones

17. Prepare to share:

- A clean draft of Sections 1–5 first (motivation, background, literature, data, strategy).
- Then a draft of Section 6 (results) once estimates are stable.

18. Make sure the structure clearly showcases your capital measurement work and updated econometric strategy.

This document should serve as the master roadmap for assembling the Econ H191 final paper out of your existing proposal, redesign memo, dossier, and the new empirical work you're completing now.